UNIVERSITY EXAMINATION BOARD

SIC

Examination Year: 2026 | Maximum Marks: **100** | Total Questions: **11** | Time Allowed: **3 Hours**

GENERAL INSTRUCTIONS:

1. All questions are compulsory unless specified otherwise.

2. Candidates must show all working, intermediate steps, and derivations to earn full marks.

3. Draw diagrams wherever necessary; label all parts clearly.

4. Figures, graphs and diagrams are provided with the respective question.

5. Use of scientific calculator is permitted unless otherwise notified.

Q.No	1	2	3	4	5	6	7	8	9	10	11	Total
Marks	25	20	20	20	15	20	15	20	20	20	20	100

Question 1 **[25 Marks]**

Find the rate of change of u(x,y) = x^2 y + x y^2 at the point (1,2) in the direction of (1,1).

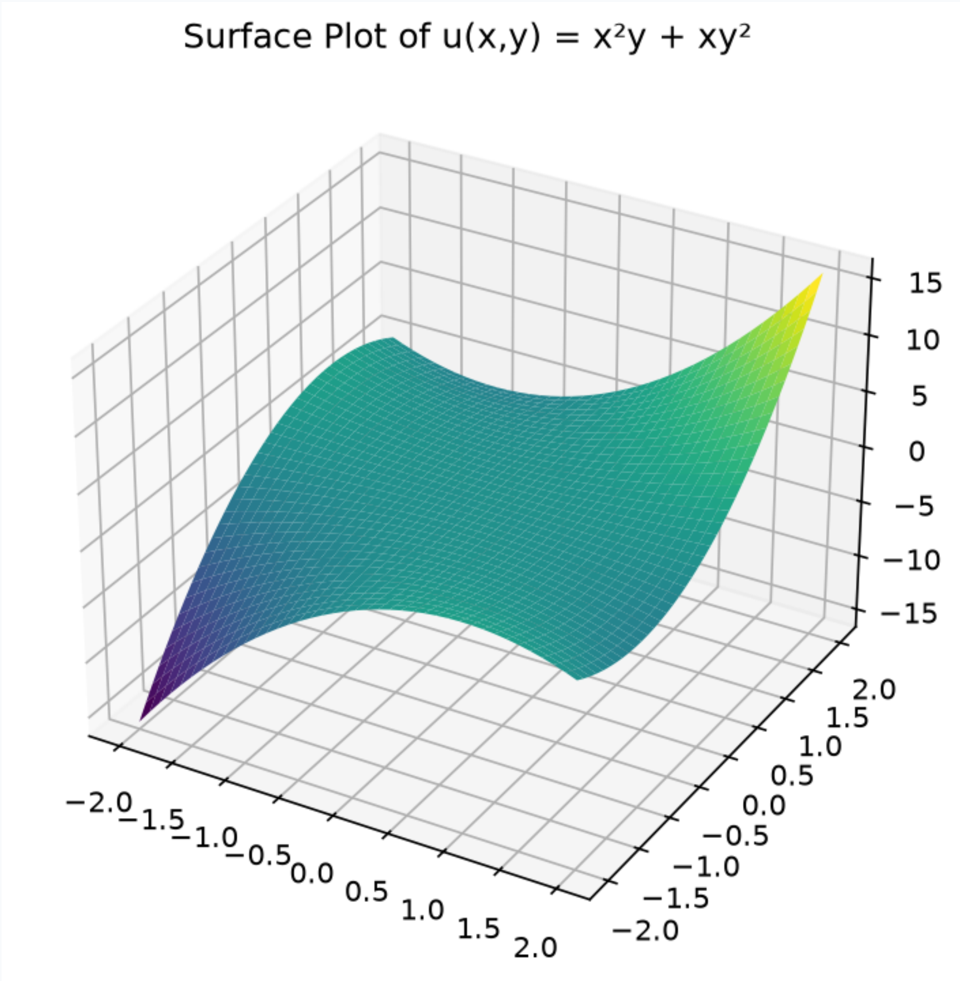

Figure 1

Question 2

[20 Marks]

Evaluate the line integral ∫_C (x^2 + y^2) dx + xy dy where C is the curve y = x^2 from (0,0) to (1,1).

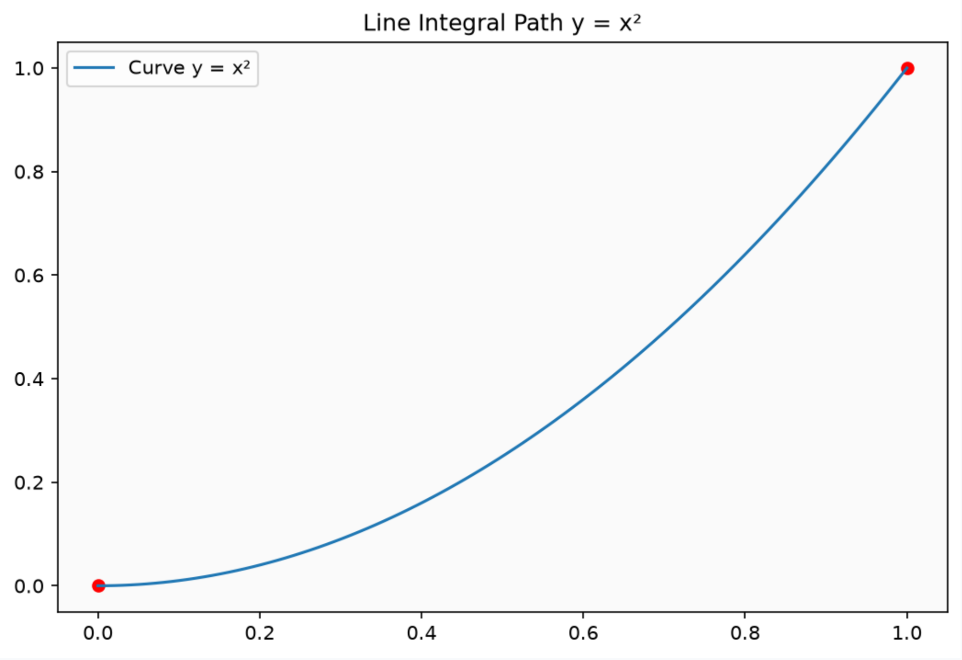

Figure 2

Question 3

[20 Marks]

Evaluate the double integral _D $(x + y)$ dA where D is the region bounded by $x^2 + y^2 = 4$ and $x + y = 2$.

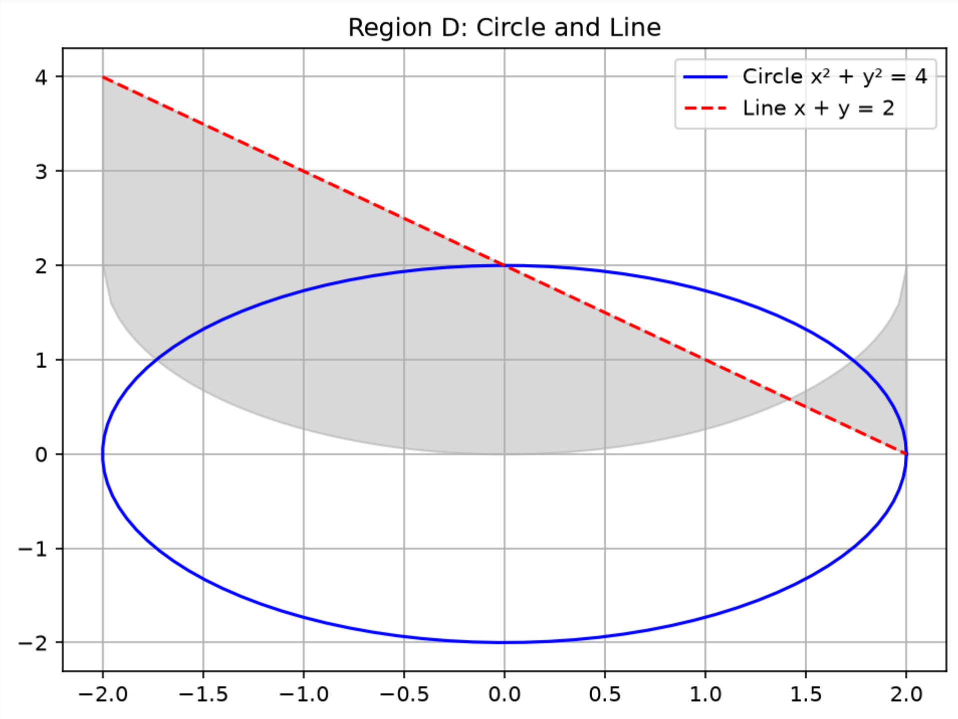

Figure 3

Question 4

[20 Marks]

Find the volume enclosed by the sphere $x^2 + y^2 + z^2 = a^2$ and the cone $x^2 + y^2 = z^2$.

[Diagram Generation Warning]

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Figure 4

Question 5

[15 Marks]

Solve the differential equation y'' + 4y = 0 with initial conditions y(0) = 1, y'(0) = 0.

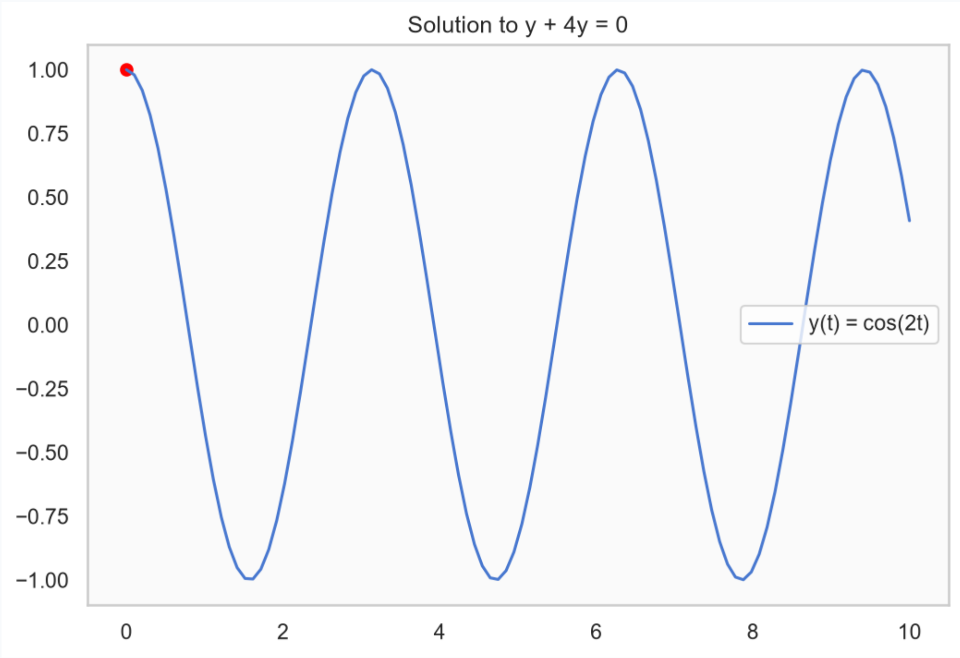

Figure 5

Question 6

[20 Marks]

Evaluate the Fourier series of f(x) = x^2 on [-π, π].

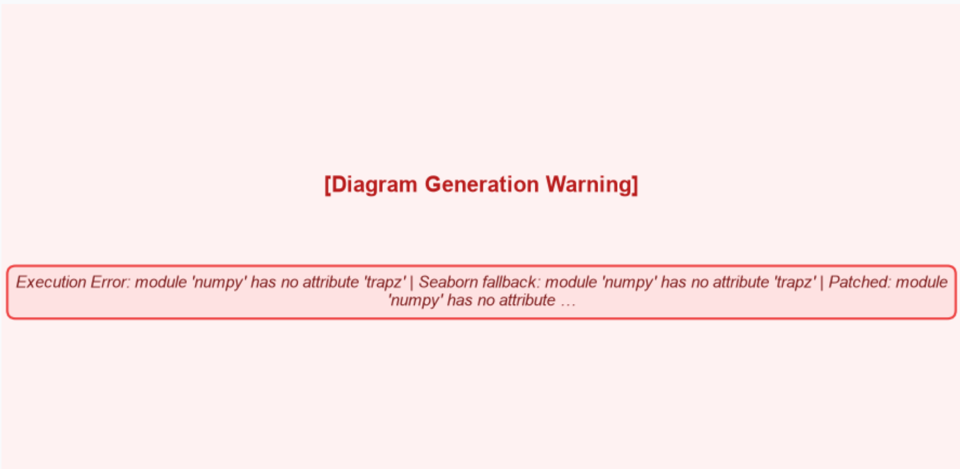

Figure 6

Question 7 **[15 Marks]**

Determine the probability density function of a random variable X with moment generating function M(t) = e^2t + 3t^2.

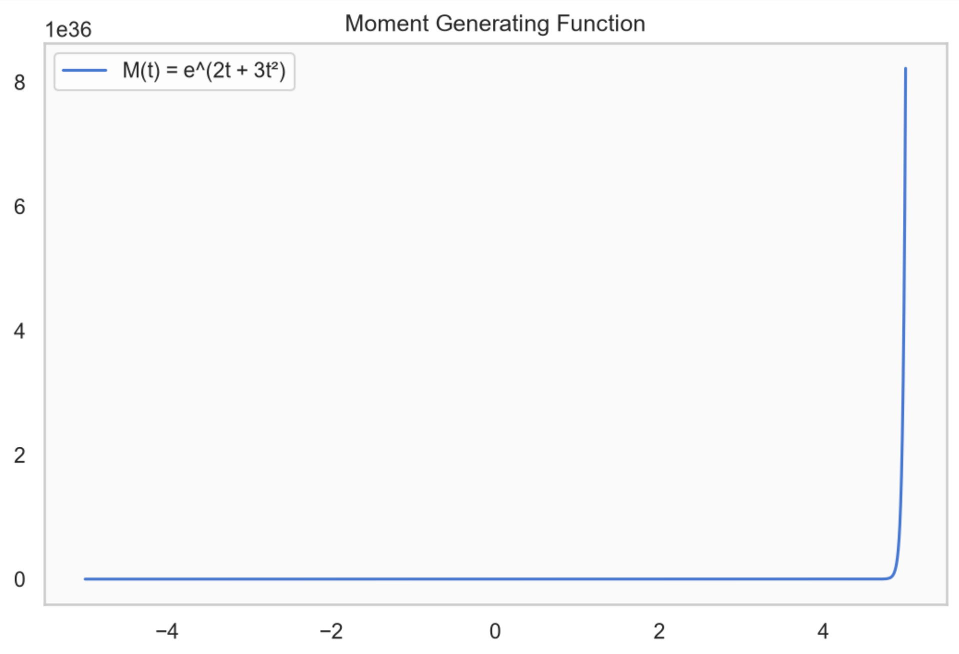

Figure 7

Question 8

[20 Marks]

Evaluate the surface integral _S (x^2 + y^2) dS where S is the hemisphere x^2 + y^2 + z^2 = a^2, z ≥ 0.

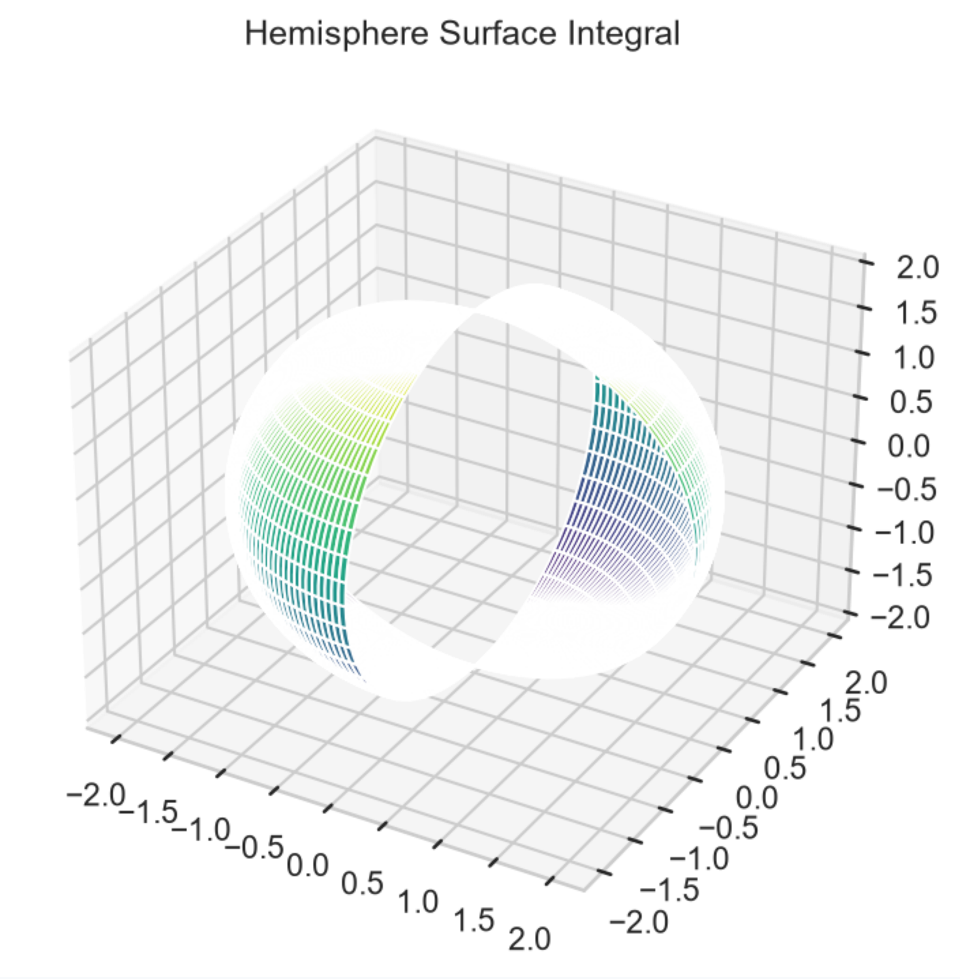

Figure 8

Question 9

[20 Marks]

Solve the partial differential equation $(\partial^2 u / \partial x^2) + (\partial^2 u / \partial y^2) = 0$ with boundary conditions $u(0,y) = 0$, $u(\pi,y) = 0$, $u(x,0) = \sin x$, $u(x,\pi) = 0$.

[Diagram Generation Warning]

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Figure 9

Question 10

[20 Marks]

Find the area enclosed by the cardioid $r = a(1 + \cos\theta)$ and the circle $r = a$.

[Diagram Generation Warning]

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Figure 10

Question 11

[20 Marks]

Evaluate the triple integral _V (x^2 + y^2 + z^2) dV where V is the region bounded by x^2 + y^2 + z^2 = 4 and z = 0.

[Diagram Generation Warning]

Execution Error: Argument Z must be 2-dimensional. | Seaborn fallback: Argument Z must be 2-dimensional. | Patched: Argument Z must be 2-dimensional.

Figure 11